

Does the Free-to-Paid Ratio Move What Spotify Pays the Labels?

Spotify Technology S.A. (NYSE: SPOT) | Gross-margin pillar, statistical evidence file

Tests whether Spotify's ratio of free users to Premium subscribers, one of the royalty targets named in its 20-F, shows up in Premium gross margin and in the labels' own revenue

-0.46

correlation of Premium GM change with ratio change, 18 quarters ($p = 0.054$)

9 of 10

model specifications with the sign the thesis predicts

0.62

correlation of Spotify's royalty-rate moves with UMG's revenue gap ($p = 0.013$)

1.73

free users per payer at Q2'27 if free growth holds at 14% (consensus 1.64)

Bottom line. The royalty mechanism is documented, and the history points the way the thesis predicts, but the statistics do not isolate it from pricing. When free users grow faster than payers, Premium gross margin expands less ($r = -0.46$ over 18 quarters). Controlling for US price increases, each +0.1 in the ratio is associated with about 81bp less Premium margin expansion ($p = 0.10$), and the sign is negative in 9 of 10 specifications; however, once the Car Thing charge and the 2024 audiobook bundle are also removed, the effect falls to 22 to 44bp and loses significance. On the label side, changes in Spotify's royalty rate show up in UMG's revenue ($r = 0.62$), which means the rate moves are real transfers rather than accounting.

Where this sits in the pitch. The pitch leads with a forward call: the Street holds free users per payer flat at about 1.65 only by assuming free-user growth nearly halves within four quarters (Section 8). This report is the supporting evidence that the ratio matters for margin, not the proof the thesis rests on. Contract mechanics will be confirmed through primary research and channel checks at the finalist stage.

Honest caveat up front. The rate term moved in Spotify's favour in 2024 and 1H26 even as the ratio rose in 1H26, so this file does not claim a significant same-quarter effect. Section 7 sets out what the evidence supports, what cuts against it, and what would change the view.

Scorecard

Twelve tests, grouped by what they ask. The right-hand column is how each result reads for the thesis, not whether the test was well built.

#	Test	Result	Reads as
Spotify side: does the ratio explain Premium gross margin?			
1	Premium GM y/y vs ratio y/y, raw (2022Q1 to 2026Q2)	$r = -0.46$, $p = 0.054$, $n = 18$	Supports
2	Same, controlling for US price steps (card-panel ATV)	-81bp per +0.1 of ratio, $p = 0.10$; R^2 rises from 0.50 to 0.59	Weak support
3	Also controlling for Car Thing (2022Q2) and the audiobook bundle (2024Q2 to 2025Q1)	-22bp (same quarter), -43bp (1Q lag), -44bp (2Q lag); $p = 0.17$ to 0.52	Not significant
4	Sign across lags 0 to 4 and both control sets	Negative in 9 of 10	Consistent
5	Subscriber growth as the driver	Adds nothing once price is controlled ($p = 0.34$)	No support
6	Churn	Not testable: no numeric disclosure since end-2021	n/a
7	Annual, 2020 to 2025	$r = -0.68$, $p = 0.14$, $n = 6$; the two largest ratio jumps (2022, 2023) were the two weakest Premium GM years	Supports, small n
Where the money moved: volume, price and rate			
8	Rate component of Premium cost growth	+EUR 112m (2022); -EUR 67m (2023, largest ratio rise); -EUR 511m (2024, ratio flat); -EUR 135m (1H26, ratio rising)	Mixed
Label side: do the labels see it?			
9	UMG growth gap to Spotify vs Spotify's cost ratio	$r = 0.62$, $p = 0.013$, $n = 15$	Rate moves are real
10	Label growth left after Spotify volume and price, vs ratio	UMG $r = 0.18$ ($p = 0.47$); WMG, approximate data, $r = 0.48$ ($p = 0.053$)	Weak
11	Label transcripts: Spotify KPIs named as a driver?	Never; Spotify is named for price increases only	No support
12	Contract language	20-F target sentence in every year FY21 to FY25; UMG CFO confirms incentive plans on growth and churn	Mechanism confirmed

How to read the statistics

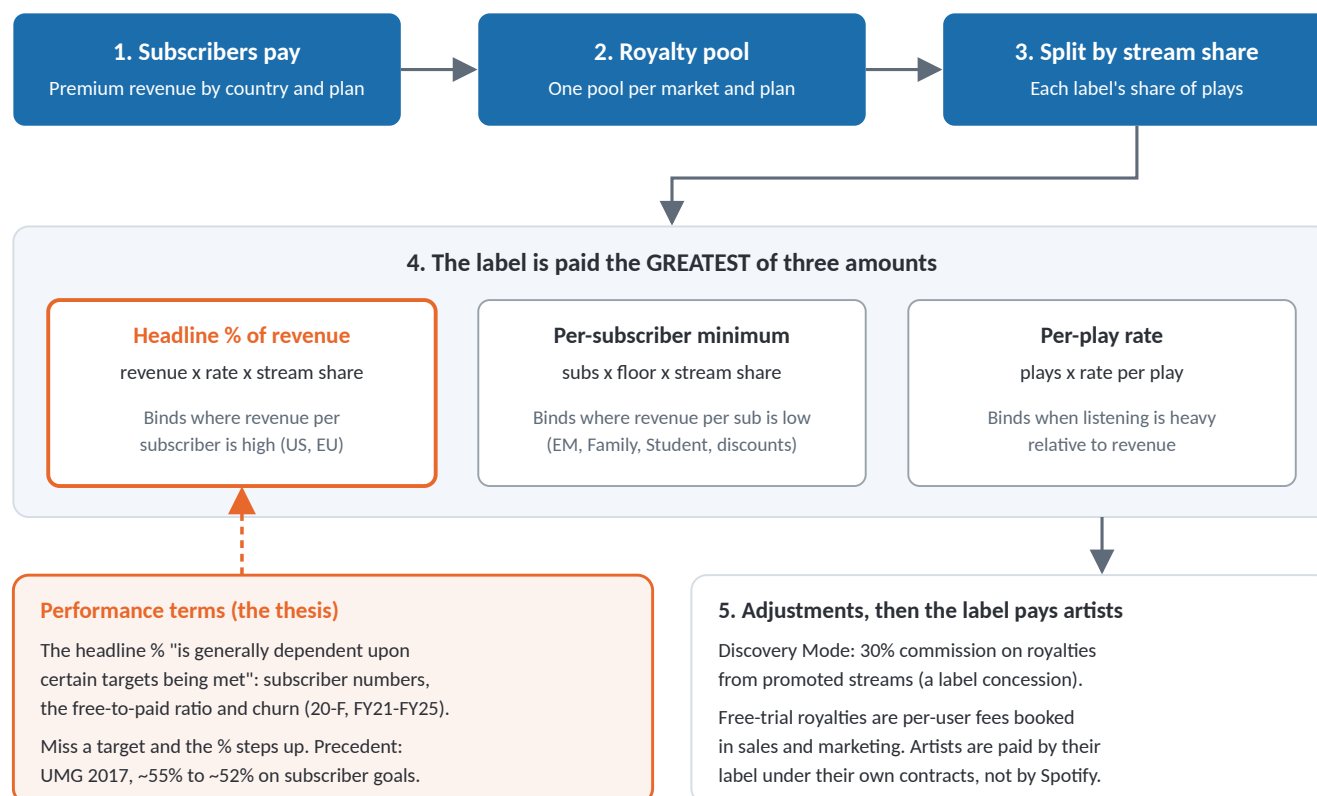
With 18 quarterly observations, only a large effect can reach conventional significance, and a contract that resets the rate in steps (recognized when the uncertainty resolves) will look like noise to a regression built to find a slope. The results are therefore best read as directional evidence: they say the history is consistent with the mechanism, not that the mechanism has been measured.

1. How the labels get paid

Labels do not take a share of all of Spotify's revenue. Revenue in each country and plan forms a royalty pool, each rights holder is paid on its own catalogue's share of streams in that pool, and each label's contract then sets its payment as the greater of several amounts (UMG describes three prongs, WMG two). Gustav Söderström, at Spotify's June 2022 Investor Day: "the business model is based on the stream share of our royalty pool."

Exhibit 1. How a euro of Premium revenue reaches a label

The performance terms (the thesis) act on the headline percentage, which is only one of the three prongs



Sources: Spotify FY25 20-F (Item 4.B, Item 5); Söderström, Investor Day, Jun 8, 2022; Muir (UMG), Nov 20, 2024; Spotify for Artists (Discovery Mode); CNBC, Apr 5, 2017.

A worked example (illustrative numbers)

Take one market with EUR 100 of Premium revenue from 10 subscribers. A major label has a 30% stream share, a 52% headline rate (the level CNBC reported for UMG in 2017, used here only for illustration) and a EUR 5 per-subscriber minimum. The highlighted cell is the prong that sets the payment.

Case	Revenue-share prong	Per-subscriber prong	Label is paid	Share of revenue
Base	100 x 52% x 30% = 15.6	10 x 5 x 30% = 15.0	15.6	15.6%
Discounts cut revenue to EUR 90	90 x 52% x 30% = 14.0	15.0 (floor binds)	15.0	16.7%
Target missed, rate steps to 55%	100 x 55% x 30% = 16.5	15.0	16.5	16.5% (+0.9pt)

Why price and margin move together

Discounts and trials raise Spotify's effective rate because the floor binds, and price increases do the reverse where the floor was binding. This is a mechanical reason, unrelated to incentives, for Premium margin to rise with price, which is why every test below controls for price.

What a missed target costs

A 3-point step on one label costs Spotify 3 points times that label's stream share, and only where the revenue-share prong binds. Across the majors and Merlin (72% of label streams) that is about 195bp of FY27 group gross margin.

What the companies have said about the terms

Term	Verbatim	Source
Stream-share pool	"the business model is based on the stream share of our royalty pool"	Söderström, Spotify Investor Day, Jun 8, 2022
Greater of three prongs	"we have a three-metric commercial model. And basically, we get paid on the higher of revenue share" and a per-play rate, "and also a per subscriber rate." Later: "it's the higher of any of these individual metrics."	Muir, UMG CFO, Nov 20, 2024
Floor = wholesale price	"When I'm thinking wholesale prices, I'm thinking our minimum subscriber rate."	Muir, Nov 20, 2024
Construct unchanged	"per subscriber minimum and the rev share. And that construct remains constant moving into Streaming 2.0."	Nash, UMG, Jul 31, 2025
Majors on similar terms	"our major DSP partners are generally all on similar economic terms in a very tight band and are generally variable deals."	Levin, WMG CFO, Nov 22, 2022
PSM escalators	"now we have PSM increases across 88% of our subscription streaming revenue ... two years ago it was zero."	Kyncl, WMG CEO, Aug 5, 2026
Performance terms	"In our agreements with certain record labels, the percentage of revenue used in the calculation of royalties is generally dependent upon certain targets being met. The targets can include such measures as the number of applicable Premium Subscribers, the ratio of Ad-Supported Users to applicable Premium Subscribers, and/or the rates of applicable Premium Subscriber churn."	Spotify FY25 20-F, Item 5 (same sentence FY21 to FY25; "applicable" added in the FY24 20-F)
Incentive plans	"we do have incentive plans embedded into those commercial agreements, which are to encourage certain behaviors or certain actions, be it to grow above a certain percentage or to reduce churn and on and on and on."	Muir, UMG, Nov 20, 2024
KPI incentives	"We do have in place ... incentive structures around hitting KPIs that might relate to margin"	Nash, UMG, May 2, 2024
Counterpoint	"it's very inappropriate for us to incentivize platforms to increase pricing."	Muir, UMG, Sep 17, 2024
Free tier is negotiated	"Because of the structure of our license agreements. The labels want a less good experience. So that people migrate to premium."	McCarthy, Spotify CFO, Feb 26, 2019
Product latitude for economics	"a quid pro quo with an economic package. The experiment didn't work as well as we hoped." The reset cost WMG \$28m to \$38m a quarter in FY22.	Cooper, WMG CEO, Sep 12, 2022 (DSP not named)

Precedent	Royalty on Universal's recordings cut "to about 52 percent, from about 55 percent," contingent on "certain subscriber goals"	CNBC, Apr 5, 2017
-----------	--	-------------------

2. Data and method

Exhibit 2. What was tested, and where each test looks



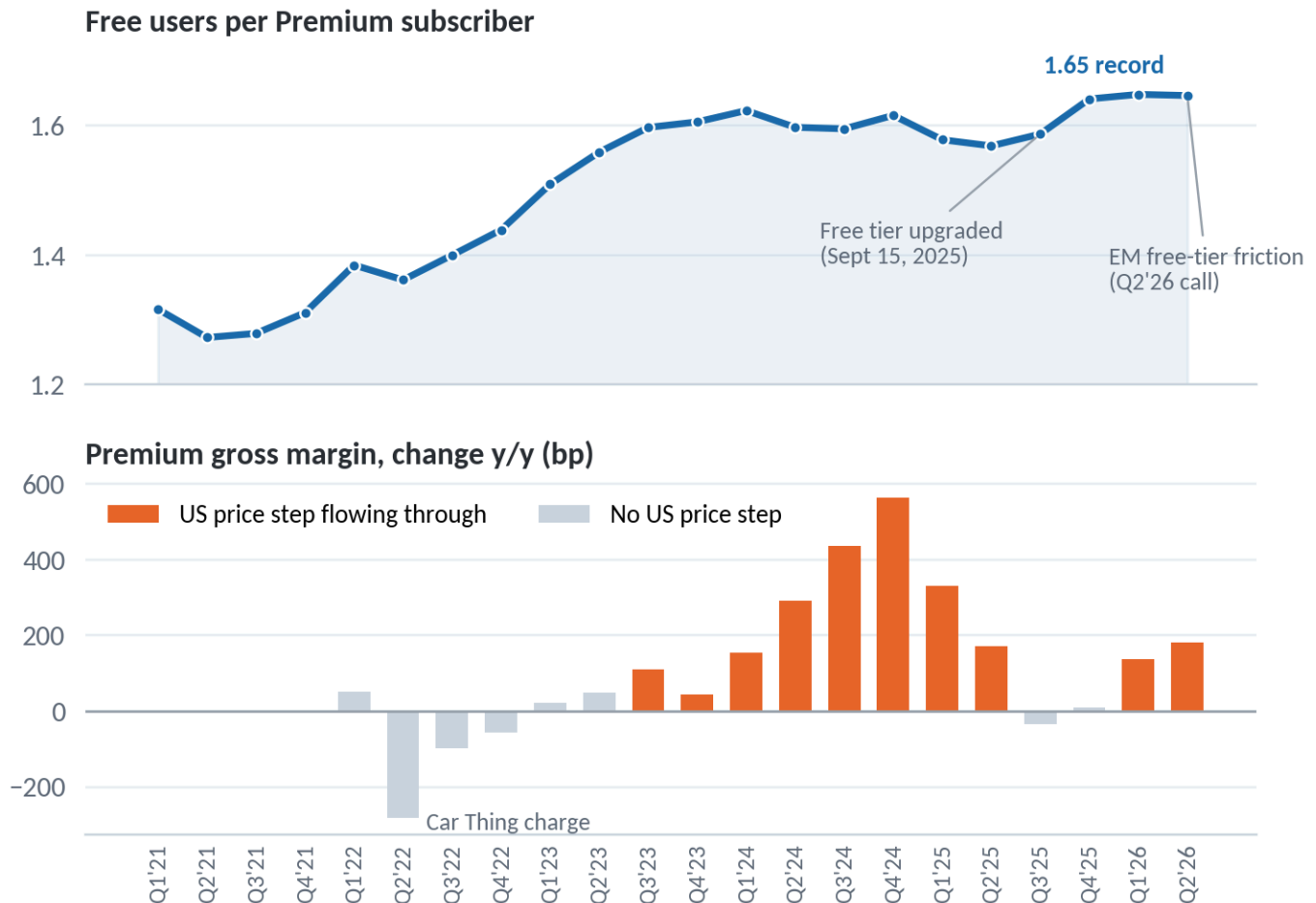
Ratio. Ad-Supported MAUs divided by Premium subscribers, both at period end. Bloomberg's "Free/Premium Ratio" row divides by the prior quarter's subscribers, so it is recomputed. **Margin.** Premium gross margin from reported Premium revenue and cost of revenue: Spotify decks and 6-Ks for 2021Q1 to 2022Q2, the Bloomberg SPOT model for 2022Q3 to 2026Q2. **Sample.** 18 y/y observations, 2022Q1 to 2026Q2. **Changes, not levels.** The ratio and Premium GM both trend up, so they correlate at +0.73 in levels; that is a shared trend, not evidence, and every test is run in changes.

Churn is not testable. Spotify's last numeric churn disclosures were 5.5% (end-2017), 5.0% (Q2'18), 4.6% (Q2'19) and 3.9% (end-2021, 2.4% in developed markets). Since then management describes churn only qualitatively, and uniformly as low; UMG's CFO in October 2023: "as the platforms are communicating, the level of churn is de minimis." The Bloomberg churn row is a forecast in every period after Q3'22.

3. Spotify side: does the ratio explain Premium gross margin?

Exhibit 3. The ratio is at a record, and Premium margin gains cluster in price-step quarters

Free users per payer rose from 1.27 (Q2'21) to 1.62 (Q1'24), moved sideways until Q3'25, then reached a record 1.65

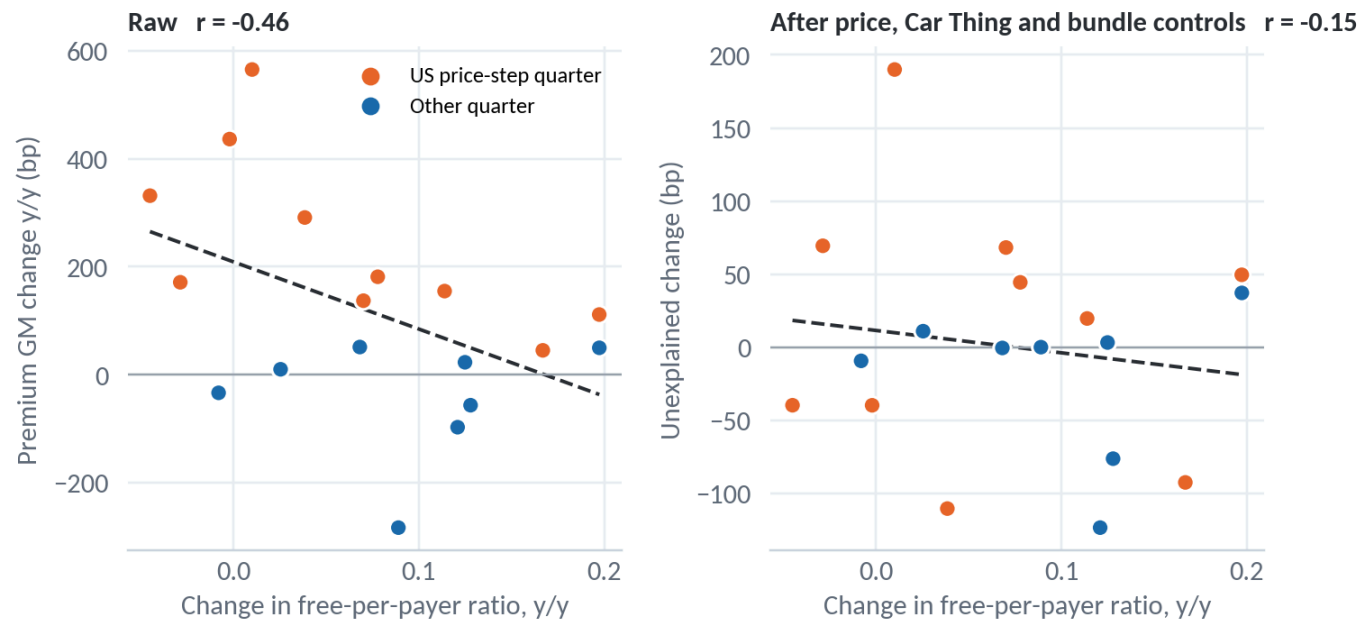


Source: Spotify decks and 6-Ks (2021 to 2Q22), Bloomberg SPOT model (3Q22 to 2Q26), Bloomberg Second Measure. Q2'22 includes the Car Thing charge.

The raw relationship is the one the thesis predicts: when free users grow faster than payers, Premium gross margin expands less ($r = -0.46$, $p = 0.054$). Most of it is price. US price steps alone explain half the variance (R^2 0.50; about 27bp of Premium GM per 1 point of card-panel ATV growth, $p = 0.001$). Adding the ratio lifts R^2 to 0.59 at -81bp per +0.1 ($p = 0.10$).

Exhibit 4. The slope survives the controls; the fit does not

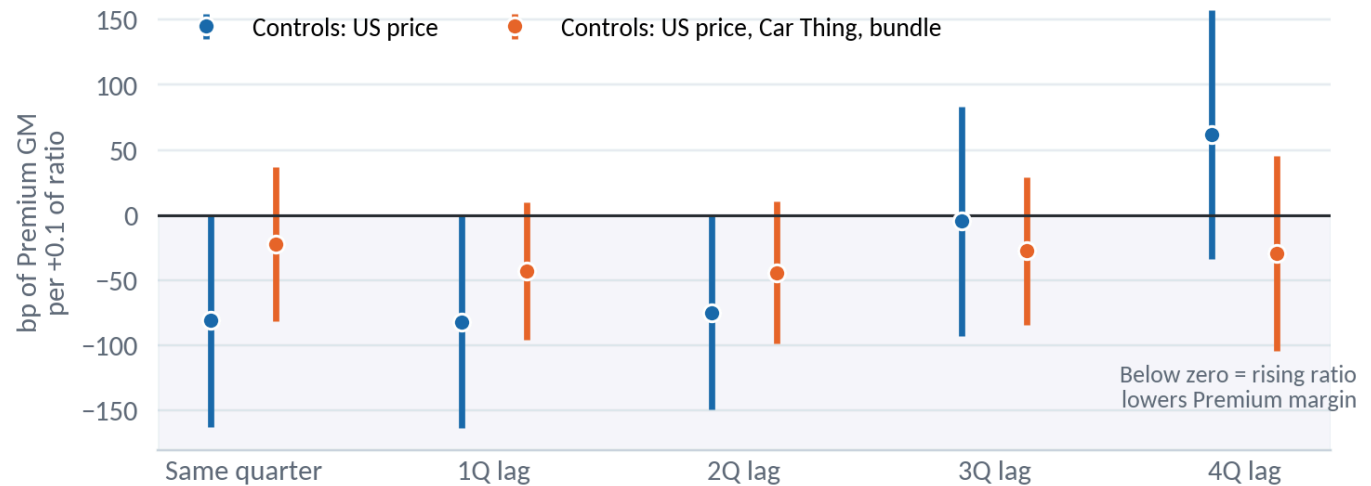
Premium GM change y/y against the change in the ratio, before and after removing price and one-offs



Orange marks US price-step quarters. Right panel: residual from a regression of Premium GM change on card-panel ATV growth and dummies for Car Thing and the bundle.

Exhibit 5. Nine of ten estimates point the same way

Estimated effect of +0.1 in the ratio on Premium GM, by lag and control set, with 90% confidence intervals

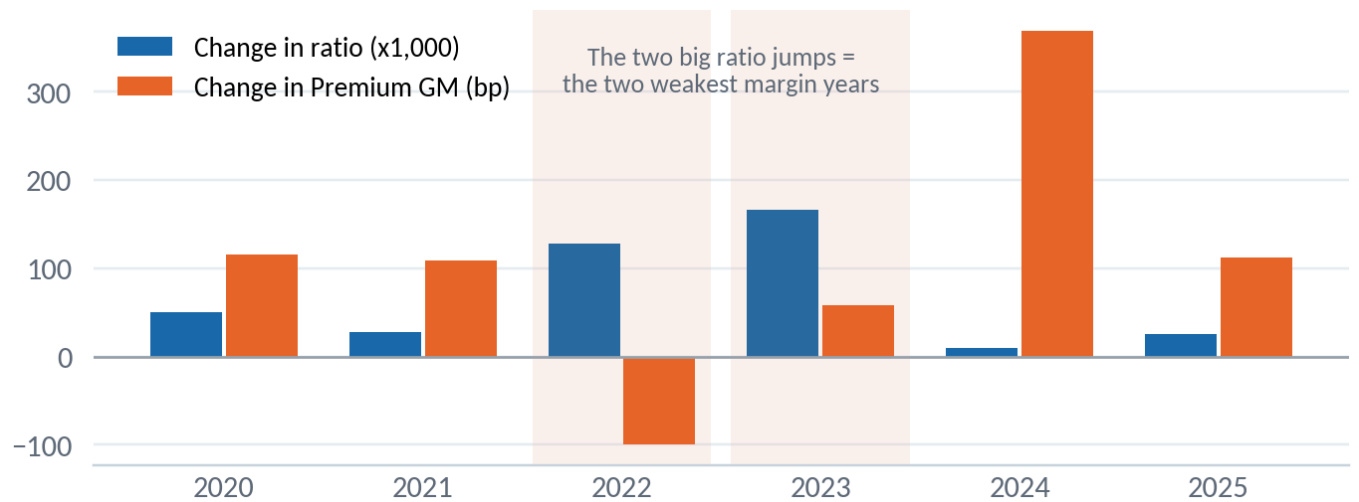


The only positive estimate is the four-quarter lag with price controls only. Adding Car Thing and bundle dummies shrinks the estimates to -22 to -44bp ($p = 0.17$ to 0.52).

Three further checks. Subscriber growth adds nothing once price is controlled ($p = 0.34$), and with it in the model the ratio coefficient is unchanged (-86bp per +0.1, $p = 0.20$). Ad-Supported gross margin is unrelated to the ratio ($r = -0.02$, $n = 12$), so whatever links the ratio to margin shows up only in Premium, which is where the 20-F places the targets. On an annual basis, the two largest ratio increases coincided with the two weakest Premium margin years.

Exhibit 6. Annual view, 2020 to 2025

$r = -0.68$ ($p = 0.14$, $n = 6$). Both weak years also carried one-off charges in cost of revenue, so they are not clean



4. Separating volume, price and rate

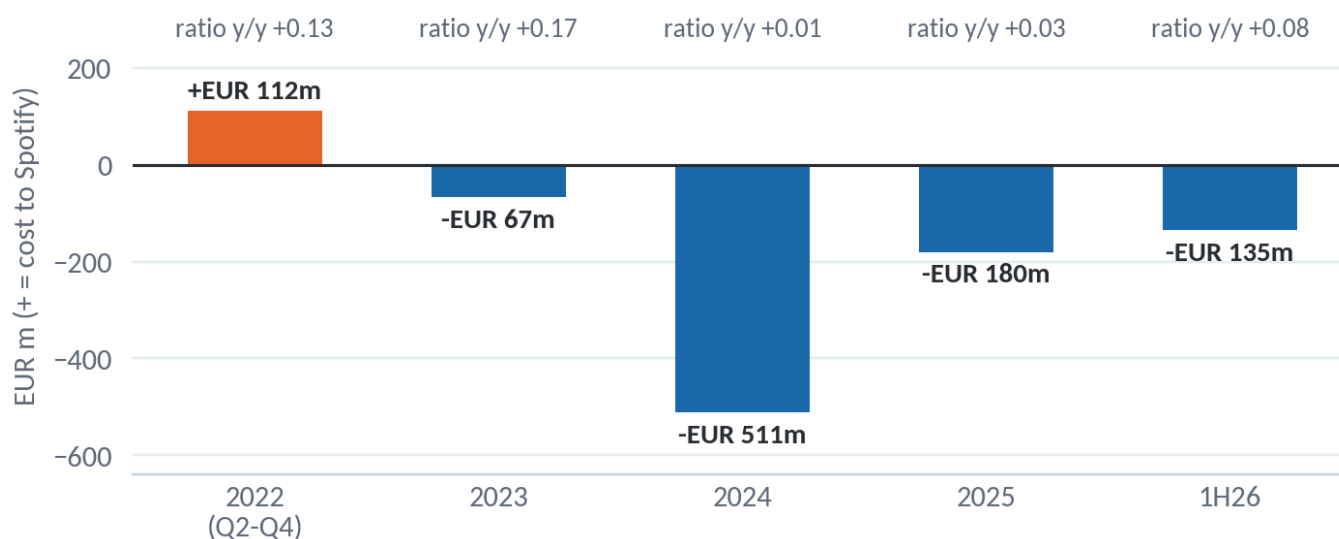
A performance incentive cannot show up as more subscribers or higher prices; it can only show up as a change in how much Spotify pays per euro of revenue. Each quarter's y/y change in Premium cost of revenue therefore splits exactly into three terms, and only the third is where the thesis can live.

Exhibit 7. Where a performance incentive can (and cannot) appear



Exhibit 8. The rate term: a cost when the ratio climbed in 2022, a large saving when it stalled in 2024

Rate component of the y/y change in Premium cost of revenue, with the change in the ratio over the same period



EUR m	Change in Premium cost of revenue	Volume	Price	Rate	Ratio y/y, period end
2022 (Q2 to Q4)	1,057	640	305	+112	+0.13
2023	876	1,156	-214	-67	+0.17
2024	1,093	1,032	571	-511	+0.01
2025	860	1,087	-47	-180	+0.03
1H26	511	471	175	-135	+0.08
Total	4,397	4,386	791	-781	

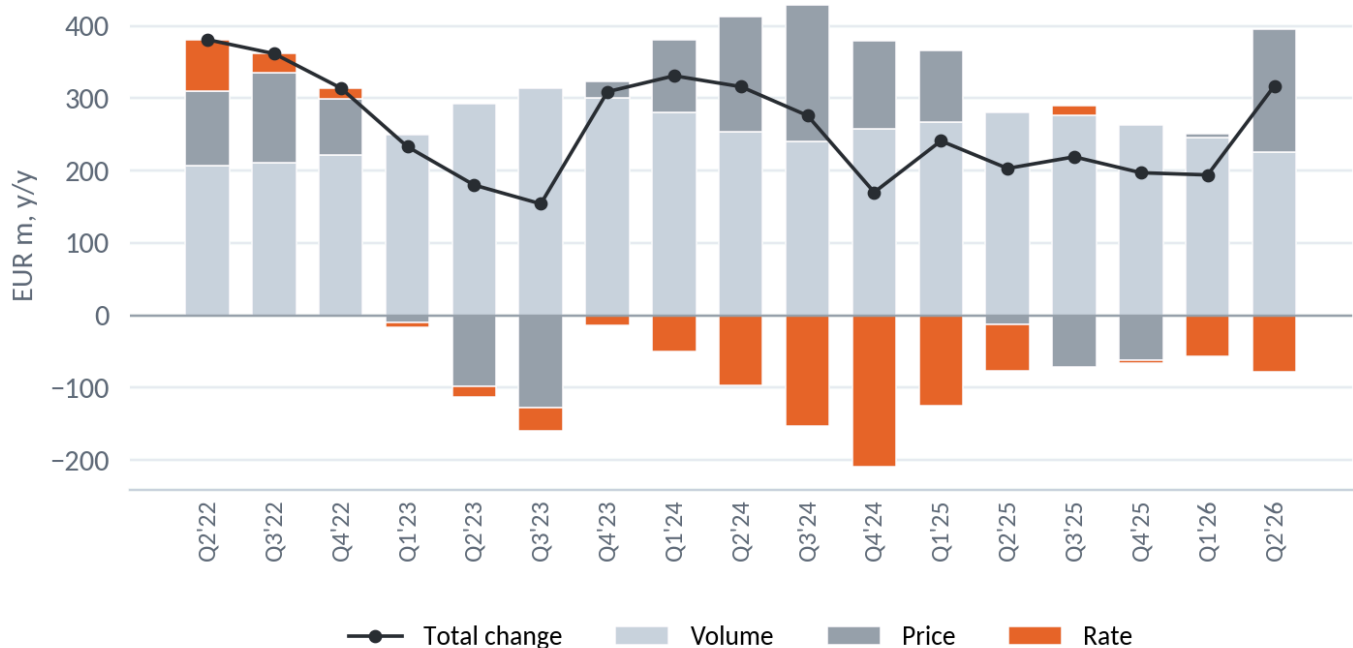
Volume explains essentially all of the EUR 4.4bn increase. Rights holders and other cost lines earned EUR 0.8bn from higher revenue per subscriber, and a lower rate handed almost exactly that amount back to Spotify. The rate term was

a cost in 2022 as the ratio climbed, roughly neutral in 2023 despite the largest ratio increase, and a EUR 511m saving in 2024 when the ratio stalled (about 370bp of 2024 Premium revenue). This shows the rate is not fixed, which is the assumption consensus makes.

Two cautions

The 2024 saving coincides with the US price step and the bundle, and management attributed every 2024 GM beat to "content cost favorability." In 1H26 the rate saved EUR 135m while the ratio rose, the opposite of what a same-quarter incentive would produce. Premium cost of revenue also includes publishing royalties, audiobooks, delivery and payment fees, so the rate term is broader than the label rate.

Exhibit 9. Quarterly split of the change in Premium cost of revenue



EUR m, y/y. Q2'22 includes the EUR 31m Car Thing charge. Revenue per subscriber is in reported euros, so FX runs through the price term.

5. Label side: do the labels see it?

If Spotify's rate moves are real, the labels should receive them. The test compares UMG's subscription growth with Spotify's Premium revenue growth: when Spotify pays more per euro, UMG's revenue should grow faster than Spotify's.

Exhibit 10. How a rate change should show up in UMG's numbers

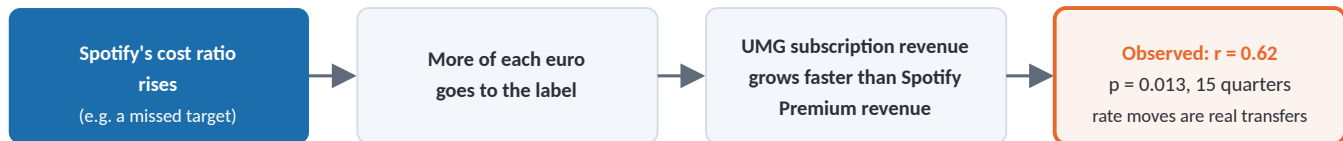
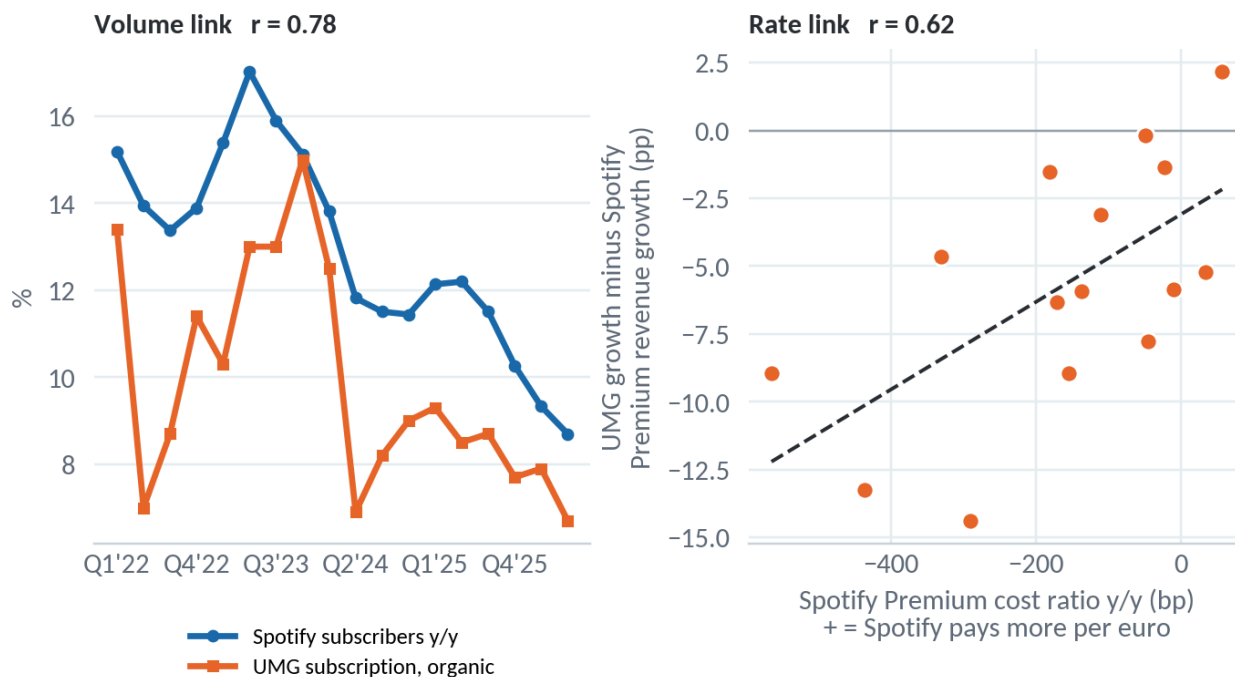


Exhibit 11. UMG tracks Spotify's volume, and its growth gap narrows when Spotify's cost ratio rises



Left: UMG organic subscription growth vs Spotify subscriber growth (2026 UMG figures ex Downtown). Right: UMG reported subscription growth minus Spotify Premium revenue growth, both in reported euros, vs the y/y change in Spotify's Premium cost ratio.

UMG's organic subscription growth tracks Spotify's subscriber growth ($r = 0.78$, $n = 18$). A volume-and-price model explains 63% of its variance, and what is left has almost no relationship with the ratio ($r = 0.18$, $p = 0.47$). WMG's quarterly subscription growth has to be pieced together from call commentary, with several quarters given as ranges and converted to midpoints; on that basis its residual correlates 0.48 with the ratio ($p = 0.053$). The sign fits, but the data are too rough to lean on.

The cleaner test is the growth gap. UMG grew more slowly than Spotify's Premium revenue in 14 of 15 quarters, and the gap narrows when Spotify's cost ratio rises ($r = 0.62$, $p = 0.013$). The gap also narrows when the ratio rises ($r = 0.45$, $p = 0.09$), a weaker result driven mainly by 2024. UMG's line includes every DSP and, from 2026, Downtown, so other partners also move the gap.

6. What the labels say about Spotify

Every passage on DSP payment terms, Spotify, subscribers, churn, free tiers and one-off payments was tagged: 601 excerpts from UMG, 688 from WMG (177 Spotify-specific) and 790 from Spotify.

- **No label ever links its results to Spotify's ratio, conversion targets or churn.** KPI terms are described only generically (Muir, Nash). WMG has never disclosed terms tied to DSP subscribers, conversion, free-to-paid ratio or churn.
- **When labels name Spotify, it is for price.** UMG Q4'23: "driven largely by the price increases at Spotify." UMG Q4'24: "the Spotify 2023 price increases anniversaried out in Q4." WMG CQ1'24: "we've now seen the full impact of the Spotify price increases in our results." WMG, Aug 7, 2024: "don't look to just one company, Spotify, namely, as the proxy for the entire industry."
- **UMG singled Spotify out as the healthy partner in Q2'24:** "While Spotify, YouTube and many regional and local platforms have continued to exhibit healthy growth in subscribers, other large partners who have been less successful in driving global adoption have seen a slowdown in new subscriber additions."
- **Label tone does not lead Spotify beats or misses.** Spotify met or beat its subscriber guide in 15 of the 16 quarters with a numeric guide (by 0 to 3m; the miss was Q1'22, -1m, on the Russia exit). WMG's weak quarters (CQ4'22, CQ1'23, CQ1'25) do not line up with Spotify misses; the first two coincide with rapid ratio increases, but WMG attributed them to the ad market, its release schedule and a prior-year extra week, and CQ1'25 to tough comparisons, "pressure in the end market", a lighter release schedule and China.
- **One possible lead.** UMG, Jul 30, 2026, on its Q2 slowdown: "the typical fluctuations in subscriber growth related to timing of promotions and price increases." Five days later Spotify guided Q3 net adds to +5m, the same as Q3'25, and flagged added free-tier friction in emerging markets.

Quarter by quarter

Qtr	Ratio	Ratio y/y	Subs vs guide (m)	GM vs guide: Spotify's stated driver	UMG tone	WMG tone
22Q1	1.385	+0.07	-1	+20bp	Strong	Strong
22Q2	1.362	+0.09	+1	Below: Car Thing (-109bp), partly offset by prior-period royalty accruals	Strong	Mixed
22Q3	1.400	+0.12	no guide	-50bp: publishing accrual on a non-US renewal	Strong	Mixed
22Q4	1.439	+0.13	no guide	+80bp: podcast spend; Marketplace	Strong	Weak
23Q1	1.510	+0.13	+3	+20 to +30bp	Strong	Weak
23Q2	1.559	+0.20	+3	In line (adjusted)	Strong	Strong
23Q3	1.597	+0.20	+2	+40bp: music favorability	Strong	Mixed
23Q4	1.606	+0.17	+1	+10bp: podcast	Spotify price rises	Strong; +\$27m DSP renewal
24Q1	1.623	+0.11	0	+120bp: content cost favorability	Strong; Spotify named	Strong; Spotify price rises
24Q2	1.598	+0.04	+1	+110bp: music content cost; Marketplace	Mixed; Spotify healthy	Strong; DSP true-up

24Q3	1.595	0.00	+1	+90bp: content cost; US publishing rate	Mixed	Strong
24Q4	1.616	+0.01	+3	+40bp: content cost favorability	Strong; EUR 20m catch-up	Mixed
25Q1	1.578	-0.05	+3	+10bp: podcast ads; content cost	Strong	Weak
25Q2	1.569	-0.03	+3	In line: Partner Program cost	Strong	Mixed
25Q3	1.587	-0.01	0	+50bp: prior-period rights-holder estimates (1H25)	Strong	Strong
25Q4	1.641	+0.03	+1	+20bp: content cost favorability	Strong	Strong
26Q1	1.648	+0.07	0	+20bp: other cost of revenue; mix	Strong	Strong
26Q2	1.647	+0.08	+1	+30bp: timing; Canada DST reversal	Mixed; promo timing	Strong

WMG quarters mapped to calendar quarters (WMG fiscal year ends Sep 30). Tone as coded from each call. Subscriber and GM guides from Spotify's prior-quarter calls.

Catch-ups and true-ups tied to DSP deals

When	Company	Item
Q2'21	UMG	EUR 41m one-time catch-up from a digital partner, in subscription revenue
FY22 (all four quarters)	WMG	\$28m to \$38m a quarter lower from resetting an above-market DSP deal (the "quid pro quo" experiment)
FQ1'24 (CQ4'23)	WMG	+\$27m upfront from an international DSP renewal
CQ2'24	WMG	DSP true-up worth about 450bp of the following year's comparison
Q4'24	UMG	EUR 20m catch-up "from a DSP partner related to new product rollouts in the second and third quarters of 2024" (partner not named; timing fits Spotify's 2024 US plan changes)
Q3'25	Spotify	GM beat "primarily driven by changes in prior period estimates for rights holder liabilities, nearly all related to the first half of 2025" (EUR 11m in Premium). 1H25 was the only stretch with a falling ratio (-0.05, -0.03 y/y) and 12% subscriber growth. Sony re-signed in the same quarter, so a renewal true-up is the alternative reading
Q2'26	UMG	About -1pp of subscription growth from "minimum guarantee accruals, audits, and catch-up payments"

7. Verdict and what would change it

What the evidence supports

- The mechanism exists: it is in every 20-F since FY21, and UMG's CFO confirmed incentive plans on growth and churn.
- The rate is not fixed: it moved by about 370bp of Premium revenue in 2024.
- Rate moves are real transfers that show up in UMG's numbers ($r = 0.62$).
- The direction is consistent: negative in 9 of 10 specifications, -81bp per $+0.1$ after price.
- The rate moved against Spotify in 2022 as the ratio climbed, and the Q3'25 rights-holder true-up related to 1H25, the one stretch in which the ratio fell.

What cuts against it

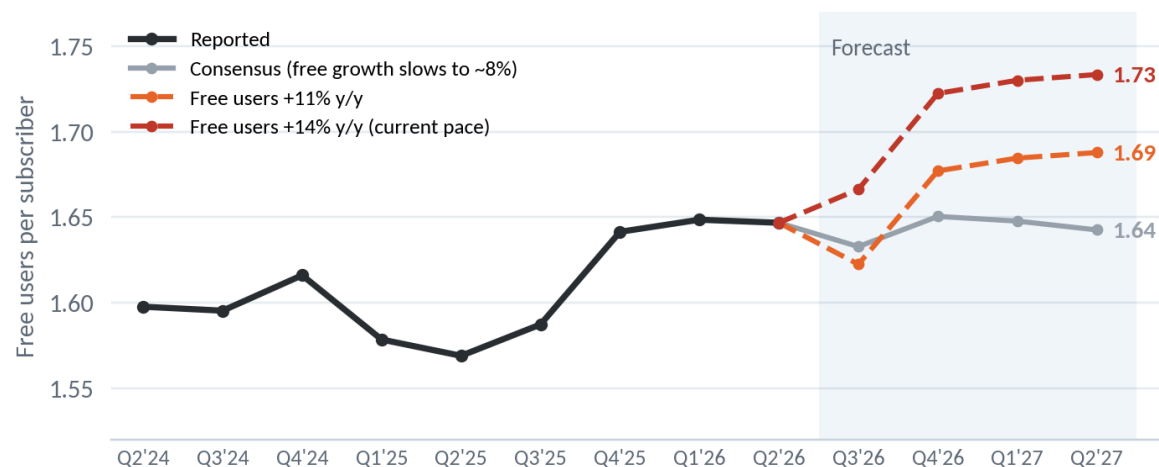
- The ratio loses significance once price and one-offs are controlled.
- In 2023, the largest ratio increase, the rate was roughly neutral; in 1H26 it fell while the ratio rose.
- Neither label mentions Spotify KPIs.
- Per-subscriber floors and the bundle explain much of 2024 without any incentive.
- The 2025 renewals may have reset the targets to levels Spotify clears easily, and thresholds, step sizes and measurement periods are not visible.

8. How this is used in the pitch

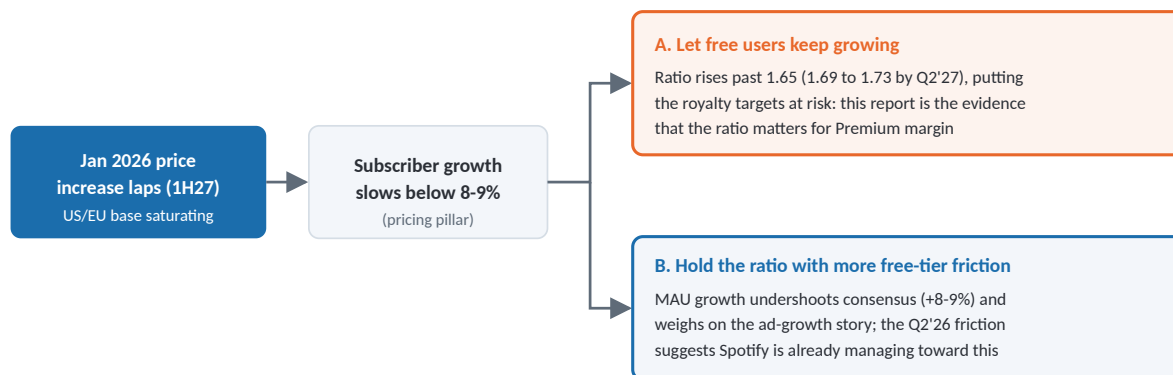
The statistics are the supporting leg, not the lead. The pitch leads with a forward call on the KPI the contracts reference: consensus holds free users per payer flat at about 1.65 only because it assumes free-user growth falls from 14% to about 8% within four quarters while subscriber growth holds at 8 to 9%. If subscriber growth slows as the January 2026 price increase laps and the US and European base saturates, Spotify faces two outcomes, and both are worse than consensus.

Exhibit 12. Consensus needs free-user growth to nearly halve for the ratio to stay flat

Free users per payer: reported, Bloomberg consensus, and two scenarios holding consensus subscribers



Source: Spotify 6-Ks; Bloomberg MODL consensus (Ad-Supported users and Premium subscribers, segment rows). Scenarios apply the stated y/y free-user growth to the year-ago quarter and divide by consensus subscribers.

Exhibit 13. Either outcome is worse than consensus**Suggested two-pager wording**

Spotify's 20-F states that the share of revenue it pays certain labels depends on subscriber growth, the ratio of free users to subscribers, and churn (a structure its former CFO attributed to labels wanting "a less good experience" on the free tier). Over 2022 to 2026, Premium gross margin has moved inversely with that ratio (roughly 80bp per 0.1 increase after controlling for price, negative in 9 of 10 specifications), and changes in Spotify's royalty rate show up in UMG's revenue from Spotify ($r = 0.62$), which indicates the rate moves are real transfers to the labels.

Do not cite: the fully controlled coefficients as significant, subscriber growth as a driver, or the UMG residual test. A judge will ask for the t-stat; "directionally consistent" is accurate.

Next steps and kill criteria

Step	What it answers
Q3'26 prints (early Nov)	Consensus: 498m free users, 305m subscribers, ratio 1.63. Free users above ~505m with subscribers at or below consensus supports outcome A; below ~490m supports B. Also any "prior period estimates" line on rights-holder liabilities, and UMG Q3 / WMG FQ4 catch-ups tied to a DSP.
Primary research (finalist stage)	Channel checks with former label business-affairs staff: are the 2025 renewals still tiered on these KPIs; step sizes and measurement periods; what "applicable" narrowed; whether trial users count as Premium subscribers.
Extend the panel to 2015	Roughly doubles n and brings the 2017 renewal, the one known rate step, into the sample: a check on whether this method can see a step we know happened.
FY26 20-F (Feb 2027)	Changes to the target sentence, aged rights-holder liabilities and the minimum-guarantee schedule.
Kill criteria	Premium GM keeps expanding through 1H27 with the ratio at or above 1.65 and no unfavourable true-ups, or primary research shows the targets were reset or removed in 2025.

Sources

Spotify: FY21 to FY25 20-Fs; Q1'22 to Q2'26 6-Ks and decks; 49 transcripts, 2018 to 2026, incl. Feb 26, 2019 conference, Jun 8, 2022 and May 21, 2026 Investor Days. UMG: 28 transcripts, Q3'21 to Q2'26, incl. Morgan Stanley TMT (Nov 20, 2024). WMG: 65 transcripts, FQ4'16 to FQ3'26, incl. Goldman Sachs Communacopia (Sep 12, 2022). Bloomberg: SPOT, UMG and WMG models; MODL consensus; Second Measure (ALTD) US card-panel data. CNBC, Apr 5, 2017. Analysis files: FreeToPaid_Analysis/ (panel_all.csv, specs.csv, decomp.csv, annual.csv).